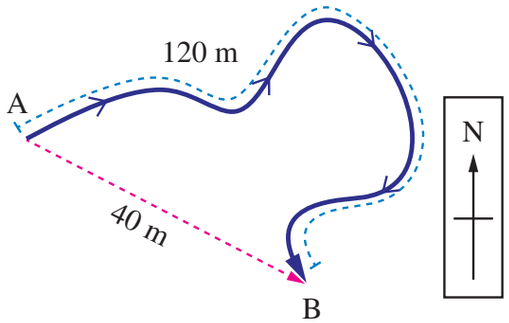
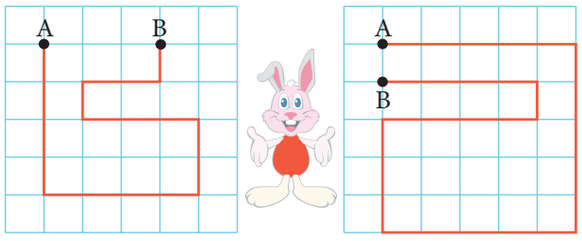
He travels 10 km along the first path. Along the second path, he travels 7 km. The distance between A and B in the case of first path is 10 km. In the case of second path, the distance is 7 km. The shortest distance between the two places is 5 km which is represented by the third path. So, the displacement is 5 km (In east direction).

The path of an object moving from point A to point B is shown in the figure. Total distance travelled by the object is 120 m. The displacement of the object is 40 m (south - east direction).



The path in which a rabbit ran is shown in the figure below. Let us consider that each square is in an unit of one square meter. The rabbit starts from point A and reaches the point B. Find the distance and displacement of it in the two figures. When will the distance and displacement be equal? (The starting point and the finishing point should be different).



Distance _____ Distance _____

Displacement _____ Displacement _____

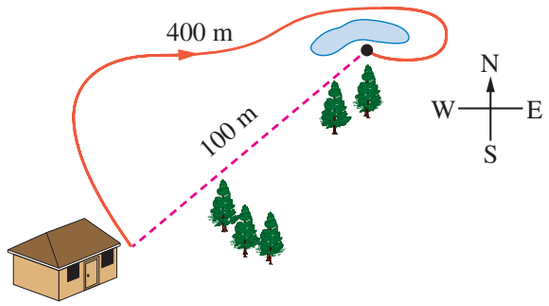
When we represent the displacement, we use a positive or negative sign depending on the direction in which it travels.



Let us consider the point A as the starting point. While the object moves from A to B the displacement is considered to be positive and it is negative, when it travels from B to A.

Subha goes to the nearby playground from her home. Look at the picture and answer the following questions.

1. What is the distance she travelled?
2. What is her displacement?



Can you answer the following questions?

- The distance travelled by an object is 15 km and its displacement is 15 km. What do you infer from this?
- The distance travelled by a person is 30 km and his displacement is 0 km. What do you infer from this?

Nautical mile

Nautical mile is the unit for measuring the distance in the field of aviation and sea transportation. One nautical mile is 1.852 km.

The unit for measuring the speed of aeroplanes and ships is knot. It means that they travel one nautical mile in one hour.

2.2 Speed - Velocity

2.2.1 Speed

In sixth standard you have already studied about speed in detail. Speed is the rate of change of distance.

Speed = Distance / Time

The unit of speed is metre/second (m/s).

We can classify speed into two types.

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